

stem cells had been compromised by prior exposure to alkylating agents.³ It is possible that high-dose cytarabine therapy is less toxic to the residual normal hematopoietic stem cells present in these patients than is combination chemotherapy with cytarabine-anthracycline antibiotics and that this provides the principal reason for the apparently greater therapeutic efficacy of high-dose cytarabine therapy in the treatment of secondary leukemia. This possibility is consistent with the observations of Gordon and co-workers, who found that the presence of leukemic cells reduced the sensitivity of normal hematopoietic cells to cytarabine by reducing the proportion of normal progenitor cells in the DNA-synthetic phase of the cycle (S phase).¹¹ Although cytarabine is S-phase specific, simultaneous exposure of cells to cytarabine and an anthracycline antibiotic kills both cycling and non-cycling cells (Preisler HD: unpublished data).

Six of the seven patients in whom chromosomal abnormalities were detected in the pretreatment marrow cells entered remission. This suggests that the previously reported prognostic importance of chromosomal abnormalities⁹ in acute nonlymphocytic leukemia may not be relevant for patients treated with high-dose cytarabine. In addition, once Patients 1 and 10 entered remission, the cytogenetic characteristics of their bone-marrow cells became normal, demonstrating that normal hematopoietic progenitor cells persist in patients with secondary acute nonlymphocytic leukemia, though they may not be detectable when the leukemia is diagnosed.

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PHYSIOLOGIC REPAIR OF AORTIC ATRESIA-HYPOPLASTIC LEFT HEART SYNDROME

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AORTIC-VALVE atresia associated with mitral atresia or stenosis, a diminutive or absent left ventricle, and severe hypoplasia of the ascending aorta and aortic arch make up a constellation of defects that was initially termed "hypoplasia of the aortic tract complexes" by Lev¹ and later called "hypoplastic left heart syndrome" by Noonan and Nadas.² The anatomic complexity of this cardiac malformation has been considered prohibitive of long survival, and generally, only supportive therapy has been recommended. Indeed, aortic atresia-hypoplastic left heart syndrome is uniformly fatal without surgical intervention, and prolonged survival after reparative surgery has not been reported. It is, however, an anomaly not easily ignored. Hypoplastic left heart syndrome is the fourth most common cardiac defect (7.5 per cent) listed by the New England Regional Infant Cardiac Program.³ According to data collected between 1969 and 1979, 223 infants were identified as having aortic atresia in New England alone. None survived for a year; their deaths accounted for nearly 25 per cent of cardiac deaths during the first week of life, as recorded by the Program. The death of such a neonate is particularly poignant since most are well-developed babies with normal birth weights, and unlike patients with other forms of congenital heart disease, they have a very low incidence of associated extracardiac anomalies.³

These compelling facts and recent advances in cardiac surgery of the infant led us, in 1979, to initiate a program of staged surgical management of neonatal hypoplastic left heart syndrome. Our preliminary results with the palliative stage have been reported.^{4,5} This paper relates our experience with the first child in this program who has been clinically well during six months of follow-up after physiologically corrective surgery for the syndrome.

CASE REPORT

The patient weighed 3.7 kg when he was born, after an uncomplicated full-term pregnancy. At 24 hours of age, a heart murmur and tachypnea were noted. No left ventricle was visualized on an echocardiogram. An infusion of prostaglandin E₁ (0.1 µg per kilogram of body weight per minute) was begun to maintain patency of the ductus arteriosus, and the child was transferred to Children's Hospital Medical Center, Boston. Physical examination on admission revealed mild cyanosis and moderate respiratory distress. The temperature was 37.7°C, the heart rate 132 per minute, the respiratory rate 88 per minute, the upper-extremity blood pressure 74/40 mm Hg, and the lower-extremity blood pressure 80/35 mm Hg. Cardiac examination revealed a hyperactive right ventricular impulse, a normal first heart sound, a single accentuated second heart sound, and a

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Table 1. Findings at Cardiac Catheterization.

SITE	AT INITIAL STUDY		AFTER PALLIATION		AFTER REPAIR	
	OXYGEN SATURATION	PRESSURE	OXYGEN SATURATION	PRESSURE	OXYGEN SATURATION	PRESSURE
	per cent	mm Hg	per cent	mm Hg	per cent	mm Hg
Superior vena cava	52	—	61	—	60	—
Right atrium	78	— (10) *	65	— (4)	55	— (14)
Pulmonary artery	93	74/38 (54)	—	— (7) †	60	18/13 (14)
Pulmonary vein	96	—	98	—	—	—
Left atrium	96	— (11)	86	— (4)	—	— (8) ‡
Right ventricle	92	74/13 —	75	94/4 —	92	95/4 —
Descending aorta	92	77/38 (56)	75	92/53 (61)	93	97/65 (80)

*Figures in parentheses are mean pressures. †Pulmonary-vein wedge pressure. ‡Pulmonary-artery wedge pressure.

Grade 3 of 6 long midsystolic murmur at the left sternal border. The liver was felt 4 cm below the right costal margin. Chest roentgenography showed moderate cardiomegaly and increased pulmonary blood flow. Electrocardiography demonstrated normal sinus rhythm (150 per minute), right-axis deviation (150 degrees), right ventricular hypertrophy, and no left ventricular forces. Findings on two-dimensional echocardiography were aortic atresia with a hypoplastic ascending aorta, mitral atresia, absence of the left ventricle, and a primum-type atrial septal defect. Cardiac catheterization demonstrated the aortic atresia-hypoplastic left heart syndrome, a hypoplastic ascending aorta (≤ 2 mm in diameter), nonrestrictive interatrial communication, and nonrestrictive patent ductus arteriosus (Table 1).

At three days of age the patient was taken to the operating room for the initial (palliative) procedure. He was placed on cardiopulmonary bypass by means of cannulation of the ductus arteriosus for arterial infusion and of the right atrial appendage for venous return. His body was cooled to 20°C, the branch vessels of the aorta were occluded, and the circulation was arrested. The distal main pulmonary artery was transected (Fig. 1). The aortic arch and ascending aorta were incised, and an anastomosis between the proximal main pulmonary artery and the ascending aorta and aortic arch was established. A 4-mm shunt of polytetrafluorethylene was established between the new ascending aorta and the branch pulmonary arteries to provide pulmonary blood flow. Cardiopulmonary bypass was reinstituted, and the patient's body warmed to 37°C. The ductus arteriosus was then ligated. Recovery was without difficulty. Arterial oxygen saturation was 85 per cent. The patient was released from the hospital on the 18th postoperative day.

The infant grew and developed in a normal fashion. At seven months of age he underwent cardiac catheterization, which revealed no appreciable pressure gradient from the right ventricle to the descending aorta, normal pulmonary venous wedge pressure, and arterial saturation of 75 per cent. By 16 months of age he had increasing cyanosis and decreasing exercise tolerance. Findings at repeat catheterization were unchanged from those noted nine months earlier (Table 1).

In April 1982 the second (reparative) procedure was performed. During deep hypothermia and circulatory arrest, the pulmonary and systemic circulations were separated by partitioning the atria so that the left atrium was associated with the tricuspid valve (Fig. 2). The right atrium was anastomosed directly to the branch pulmonary arteries, and the shunt was removed. The immediate postoperative course was uneventful. The patient was extubated 36 hours after surgery. He required right thoracotomy for moderate fluid accumulation on the 10th postoperative day. Cardiac catheterization, performed 14 days after surgery, documented a hemodynamic result comparable to that in patients with a good long-term prognosis after Fontan's operation for other forms of congenital heart disease (Table 1). The patient was released from the hospital 21 days after surgery. He was clinically well at the time of final submission of this paper, nearly six months after the operation.

DISCUSSION

The term "hypoplastic left heart syndrome" has been used to describe various degrees of underdevelop-

ment of left heart structures. To most practitioners, the central anatomic feature is absence or atresia of the aortic valve, which is always associated with severe hypoplasia of the ascending aorta and various degrees of mitral-valve and left ventricular hypoplasia. Moodie reviewed the pathological anatomy in 172 cases of aortic atresia.⁶ Fifty-seven patients (33 per cent) had associated mitral atresia and no discernible left ventricle; 103 (60 per cent) had mitral-valve hypoplasia, an intact ventricular septum, and a diminutive left ventricle; and 12 (7 per cent) had an associated large ventricular septal defect and only mild hypoplasia of the mitral valve and left ventricle. To date, no satisfying hypothesis has been advanced regarding the cause of this condition.

Perinatal survival of neonates with hypoplastic left heart syndrome is dependent on maintenance of the systemic circulation through a patent ductus arteriosus. In the majority of those with an intact ventricular septum, pulmonary venous return to the right ventricle is of necessity through an interatrial communication (usually a patent foramen ovale). Death in the first few days of life is generally due to closure of the ductus arteriosus and the consequent decrease in systemic perfusion. If the ductus arteriosus remains patent, death in infancy generally results from severe congestive heart failure and failure to thrive, because of ever-increasing pulmonary blood flow. A rare patient will fortuitously have a persistent patent ductus arteriosus supporting the systemic circulation, in combination with relatively low pulmonary blood flow (due either to pulmonary arteriolar constriction or a restrictive interatrial communication), which allows survival beyond infancy. Although one patient with aortic atresia survived to the age of 3½ years before dying of pulmonary vascular obstructive disease, the reported mean survival time ranges from 4 to 23 days.⁶

Medical treatment with anticongestive measures does little to alter the natural history of this condition. Constant infusion of prostaglandin E₁ inhibits physiologic closure of the ductus arteriosus and is a potent means of resuscitating the acidotic neonate with hypoplastic left heart syndrome. Its use, however, seems warranted only when more definitive treatment is available. The reported surgical experience is for the most part limited to case reports. Cayler et al. reported using a Waterston shunt in one infant with critical aortic stenosis,⁷ and Freedom et al. constructed a Potts shunt and banded the individual pulmonary arteries in a child with aortic atresia.⁸ Both patients died without more definitive surgery. Litwin and colleagues suggested that the considerable limitations inherent in banding the branch pulmonary arteries in the neonate could be avoided by placing a conduit from the main pulmonary artery to the descending aorta, placing a band on the main pulmonary artery distal to the con-

duit, and ligating the ductus arteriosus.⁹ Although others have recently reported the use of this concept in two infants,^{10,11} an attempt at such palliation in the first neonate in our series convinced us that the main pulmonary artery was too short in these neonates to accommodate both a conduit and a band without compromising the right pulmonary artery.⁴ Doty et al. reported palliative treatment by placement of a conduit from the main pulmonary artery to the aortic arch, ligation of the ductus arteriosus, and restriction of pulmonary blood flow with a perforated patch in the main pulmonary artery.¹² Their patient died with fibrotic narrowing of the perforation.

Principally because of the developmental work of Fontan, during the 1970s reparative surgery proved possible for patients with tricuspid atresia, another congenital cardiac malformation with only one effective ventricle — in this case, a left ventricle.¹³ This “physiologically corrective” operation is predicated on the observation that if pulmonary vascular resistance is low, as it is in the normal mature lung, circulation of blood through the pulmonary vasculature can be achieved without a pulmonary ventricle. The right atrium can serve this function. Physiologic correction in patients with tricuspid atresia is accomplished by closing the atrial septal defect and connecting the right atrium directly to the pulmonary arteries. In principle, patients with hypoplastic left heart syndrome can be similarly treated by using the right atrium as the pumping chamber for the pulmonary circulation and the right ventricle as the systemic pumping chamber. The neonate, however, normally has high pulmonary vascular resistance, which decreases to adult levels only after the first weeks of life. Thus, surgery of the type performed by Fontan for tricuspid atresia is precluded in the neonatal period,¹³ precisely when the majority of patients with hypoplastic left heart syndrome become symptomatic and die. Moreover, if pulmonary blood flow and pressure remain elevated, as in the rare infant who survives with the syndrome untreated, the pulmonary vasculature does not mature in a normal fashion.¹⁴ Instead, irreversible pulmonary-artery hypertension ensues, which precludes physiologically corrective surgery at any time. Therefore, a staged surgical approach is necessary to change the natural history of the syndrome.

In 1979 we initiated a program to evaluate the feasibility and efficacy of staged surgical management of

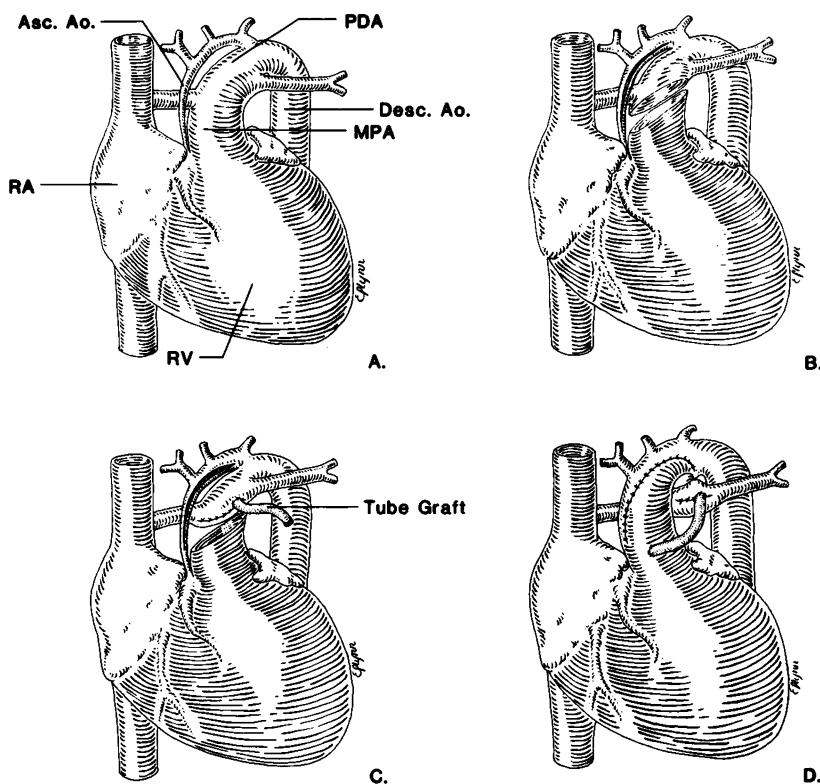


Figure 1. Procedure for Palliation of Hypoplastic Left Heart Syndrome.

The heart with aortic atresia has a diminutive ascending aorta and aortic arch and a large patent ductus arteriosus (A) (Asc. ao. denotes ascending aorta, Desc. ao. descending aorta, MPA main pulmonary artery, PDA patent ductus arteriosus, RA right atrium, and RV right ventricle). The main pulmonary artery is transected, and an incision is made in the ascending aorta and aortic arch (B). The distal main pulmonary artery is oversewn, and the distal anastomosis of the 4-mm shunt established (C). The main pulmonary artery is anastomosed to the ascending aorta and aortic arch, the ductus arteriosus is ligated, and the shunt is completed (D).

the hypoplastic left heart syndrome. On the basis of the cardiac anatomy of infants with the syndrome and the physiologic considerations for reparative surgery, the following objectives for the first surgical stage appeared essential at the outset: (1) establishment of a permanent unobstructed communication between the right ventricle and aorta, (2) limitation of pulmonary blood flow and pressure to nearly normal levels, and (3) establishment of a large interatrial communication. The first objective was necessary for adequate systemic perfusion and preservation of ventricular function, and the second two were essential for normal development of the pulmonary vasculature. The development of the operative procedures employing these principles for palliation, as well as the initial results in the first 16 neonates of our series, has been previously described.^{4,5} With this limited experience, our preferred method for palliation consists at present of reconstruction of the diminutive ascending aorta and aortic arch by anastomosis of the transected proximal main pulmonary artery to the ascending aorta and aortic arch (Fig. 1). This procedure establishes unobstructed antegrade flow from the right ventricle (the new systemic pumping chamber) through the aorta. Moreover, the use of autologous valvular and vascular tissue in the systemic circulation maximizes both technical

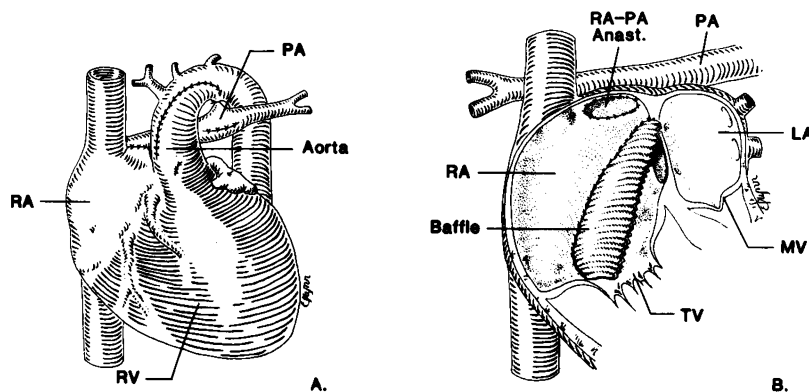


Figure 2. Reparative Procedure.

The shunt is removed and an anastomosis is made (A) between the right atrium and pulmonary arteries (PA). The interatrial baffle connects the left atrium (LA) to the right ventricle via the atrial septal defect (B) (MV denotes mitral valve, and TV tricuspid valve).

simplicity and the potential for growth and development of the reconstructed heart and great vessels. Pulmonary blood flow appropriate for normal development of the pulmonary vasculature can then be achieved either by a 4-mm central shunt, as in the patient described above, or by a conventional Blalock-Taussig shunt, which we now prefer.

Physiologic correction was completed in our patient at the age of 16 months, when the pulmonary vascular resistance had reached normal low levels. After removal of the shunt, an interatrial baffle was constructed to provide continuity between the left atrium and the tricuspid valve through the atrial septal defect, thereby separating the pulmonary and systemic circulations (Fig. 2). The pulmonary arterial circulation was completed by anastomosis of the right atrium to the pulmonary arteries. Postoperative cardiac catheterization revealed a mean right atrial pressure of 11 to 14 mm Hg, a well-functioning right ventricle with normal cardiac output, and normal systemic arterial oxygen saturation. Obviously, determination of the efficacy of such management must await long-term follow-up of a substantial group of patients. However, this child's gratifying postoperative course and current well-being serve to document the feasibility of surgically altering the prognosis of newborns with aortic atresia and other forms of hypoplastic left heart syndrome.

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STIMULATED HUMAN PHAGOCYTES PRODUCE CYTOGENETIC CHANGES IN CULTURED MAMMALIAN CELLS

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CHRONIC inflammation can be associated with cancer.¹ Phagocytes in inflammatory lesions enzymatically reduce oxygen to reactive metabolites,

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